

Observation. For a grammar in Chomsky normal form, any parse tree for a long string should have a long path.

Any long path should have at least two occurrences of some nonterminal symbol.

For a grammar in CNF - the number of symbols can at most double going down a level in the parse tree - RHS of each production contains at most 2 symbols.

We have 1 symbol at level 0
at most 2 symbols at level 1.

$\vdots$
 2^i symbols at level i .

To have 2^n symbols at the bottom level, the tree must be of depth* at least n . Thus the parse tree has at least $n+1$ levels.

* Depth - number of edges in the longest path from the root to a leaf node.

Pumping Lemma for CFLs.

if $A \subseteq \Sigma^*$ is a CFL then there exist $k \geq 0$

such that for every $z \in A$ of length at least k

can be split into five substrings $z = uvwx^i y$

such that $vx \neq \epsilon$, $|vwx| \leq k$ and for all $i \geq 0$,
 $uv^iwx^iy \in A$.

Proof.

Let G be a grammar for A in CNF.

Take $k = 2^{n+1}$, where n is the number of nonterminals of G .

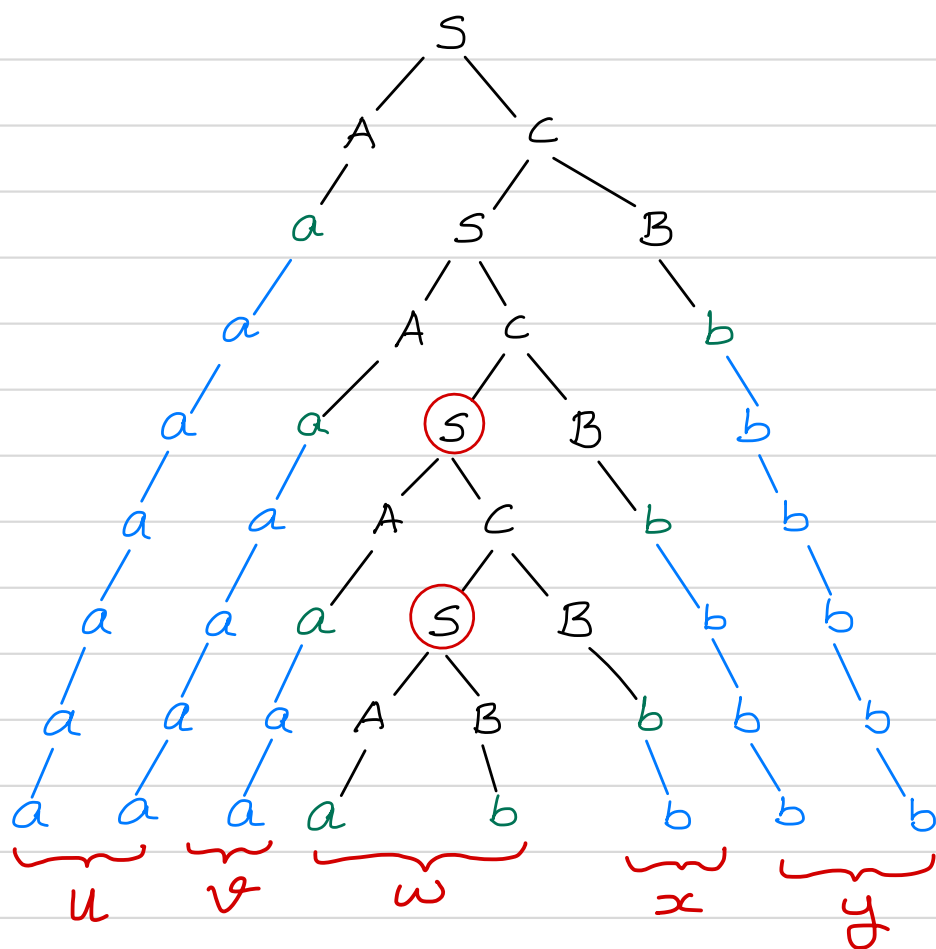
Suppose $z \in A$ and $|z| \geq k$.

Any parse tree in G for z must be of depth at least $n+1$ (i.e. there are $n+2$ levels).

Consider the longest path in the tree (it is of length at least $n+1$).

The longest path contains at least $n+1$ occurrence of nonterminals.

This implies: Some nonterminal occurs more than once.

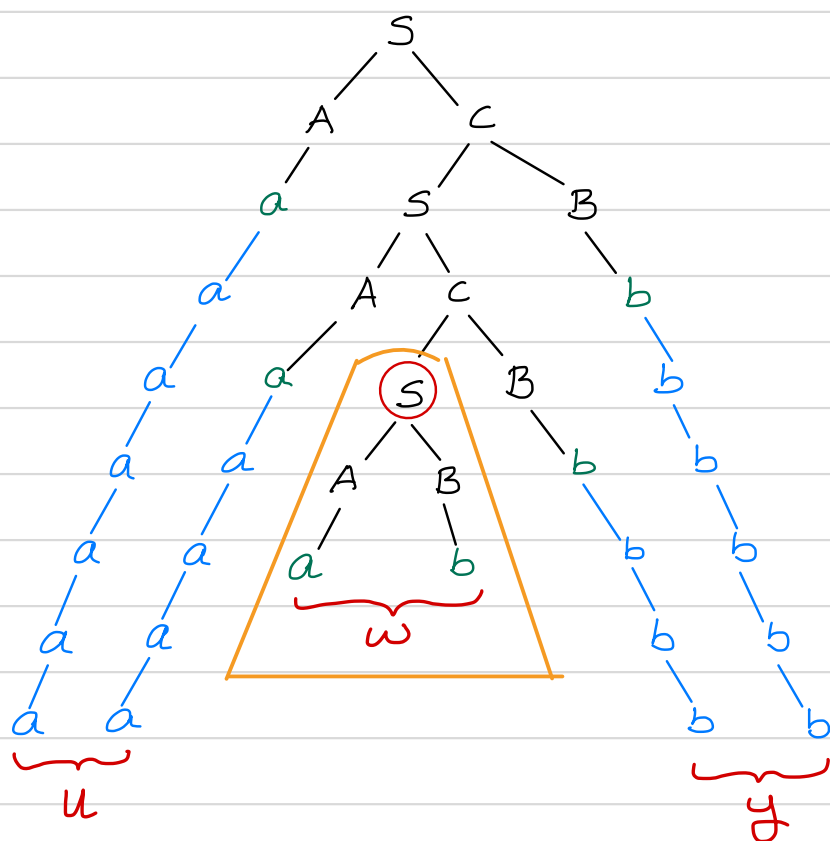
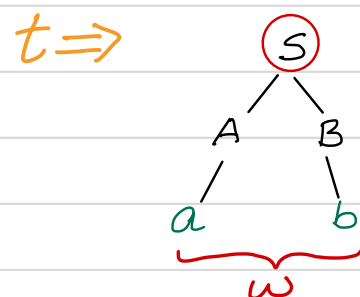
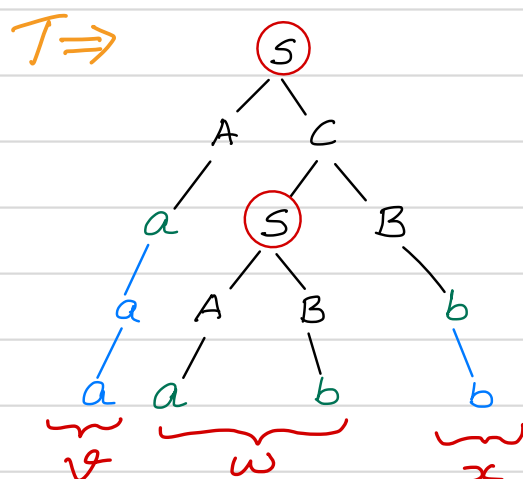
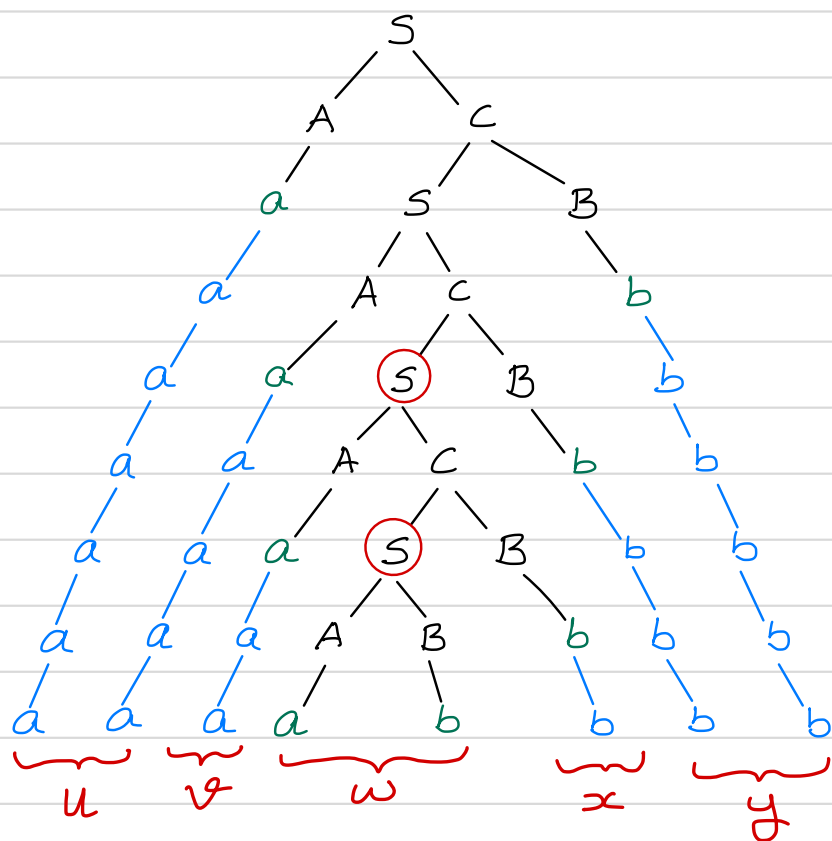


Take the first pair of occurrences of the same nonterminal along the path - traversing bottom to top.

Suppose X is the nonterminal with two occurrences.
Split $Z = uvwxy$ such that.

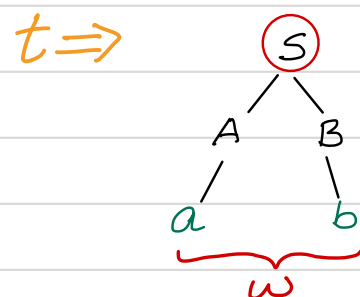
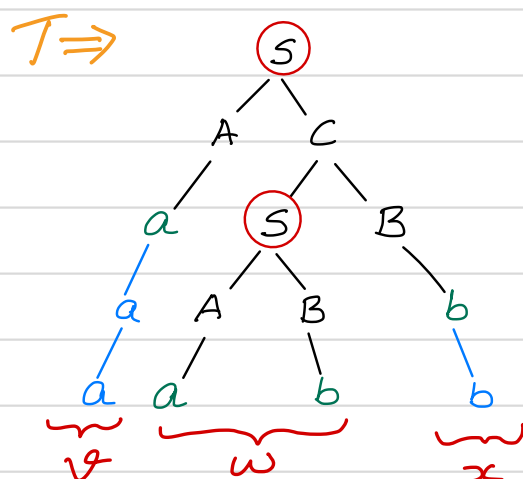
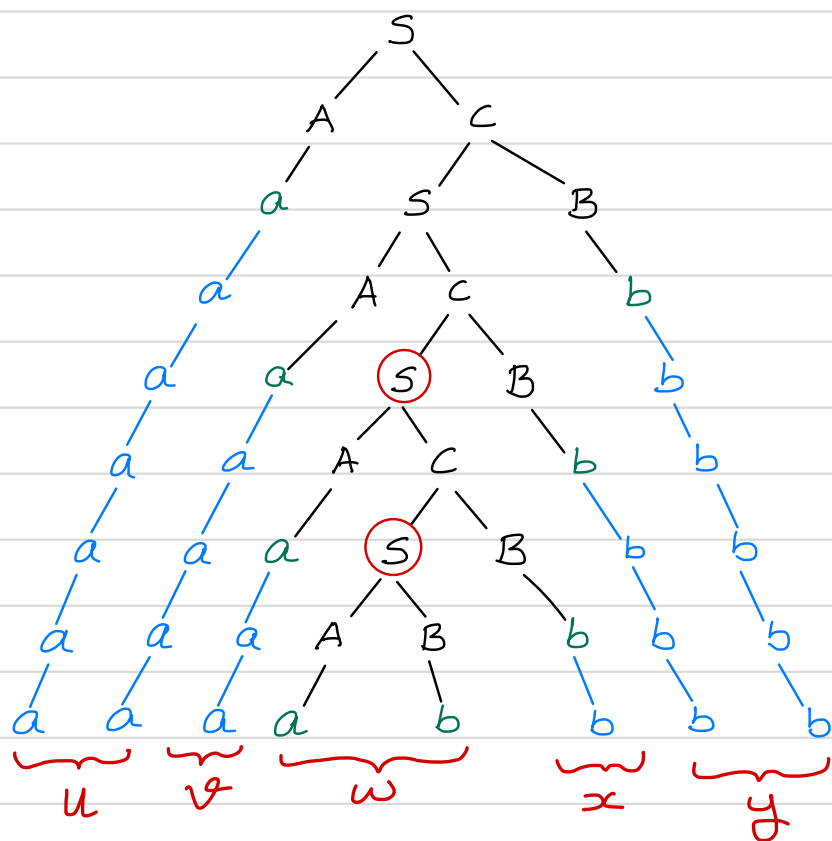
w -String (of terminals) generated by lower occurrence of x
 vwx -String generated by the upper occurrence of x

Let T - Subtree rooted at upper occurrence of x .
 t - Subtree rooted at the lower occurrence of x



Consider the string $u v^2 w x^2 y$

We can cut T and replace it with t to get a parse tree for $u v^0 w x^0 y = u w y$.



Note.

1. $\forall x \neq \epsilon$. - v and x are not both ϵ .
2. $|vwxc| \leq k$ - Since we chose the first repeated occurrence of a nonterminal from the bottom.
Depth of the subtree under the upper occurrence of the repeated nonterminal x is at most $n+1$
 $\therefore$ it can have at most $2^{n+1} = k$ terminals

□

To show that a set is not a CFL - use pumping lemma in its contrapositive form.

For all $k \geq 0$, $\exists z \in A$ s.t. $|z| \geq k$ and for all split of z into substrings $z = uvwxy$ with $v \neq \epsilon$ and $|vwx| \leq k$, there exists an $i \geq 0$ s.t. $uv^iwx^iy \notin A$

$A = \{a^n b^n a^n \mid n \geq 0\}$ is not context free.

Proof. Given k , let $z = a^k b^k a^k$. We have $z \in A$, $|z| = 3k$

Now consider any split $z = uvwxy$ with $v \neq \epsilon$ and $|vwx| \leq k$. Let $i = 2$. Consider the string uv^2wx^2y .

Case 1. v or x contains at least one "a" and at least one "b".

Case 2. v and x contains only a's.

Case 3. v and x contains only b's.

Case 4. One of v or x contains only a's and the other only b's.

To show that a set is not a CFL - use pumping lemma in its contrapositive form.

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Case 1. v or x contains at least one "a" and at least one "b".
Then uv^2wx^2y is not of the form $a^*b^*a^*$

Case 2. v and x contains only a's.
Then uv^2wx^2y has more a's than b's.

Case 3. v and x contains only b's. Then the number of b's is greater than the number of a's.

Case 4. One of v or x contains only a's and the other only b's. Then uv^2wx^2y is not of the form $a^m b^m a^m$.

Thus in all cases, the resulting string $uv^2wx^2y \notin A$.

By pumping lemma, A is not a CFL.

$$A = \{\omega \mid \#_a(\omega) < \#_b(\omega) < \#_c(\omega)\}$$

Given k , choose $z = a^k b^{k+1} c^{k+2}$; $z \in A$.

Consider any split of $z = uvxyz$; $|vxy| \leq k$
 $vy \neq \epsilon$.

Cases.

- v contains only a 's and y contains only b 's.

So $v = a^m$ and $y = b^l$; $1 \leq m+l \leq k$

Take $i=3$.

So the string $uv^3xy^3z = a^{k+2m} b^{k+2l+1} c^{k+2}$

Since $vy \neq \epsilon$, either $m \geq 1$ or $l \geq 1$.

- if $m \geq 1$ then $k+2m \geq k+2$

- if $l \geq 1$ then $k+2l+1 > k+2$

Thus $a^{k+2m} b^{k+2l+1} c^{k+2} \notin A$.

- v contains only a 's; y contains a 's and b 's.

- v contains a 's and b 's; y contains only b 's.

- v and y contains only a 's; $a^{k+m} b^{k+1} c^{k+2} \notin A$ $m \geq 1$

- v and y contains only b 's; $a^k b^{k+m+1} c^{k+2} \notin A$ $m \geq 1$

- v and y is in $b^{m+1} c^{m+2}$